FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Common Endocrine Disorders

Tutorial | Week 6 · Tetanus, skin, poisons & hormones | Paediatrics | Dr Namugga

The paediatric endocrine conditions a Ugandan junior clerk must recognise are type 1 diabetes (often first diagnosed in diabetic ketoacidosis, which needs careful weight-based fluids, low-dose insulin and potassium per ISPAD 2022), congenital hypothyroidism (treatable cause of intellectual disability, missed because Uganda has no routine newborn screening), short stature (mostly nutritional stunting, but sometimes endocrine), ambiguous genitalia from congenital adrenal hyperplasia (a salt-wasting emergency), and nutritional rickets.

Must know before the ward round

1. Diabetes in a child: random glucose ≥11.1 mmol/L with symptoms (polyuria, polydipsia, weight loss), or fasting ≥7.0 mmol/L, or HbA1c ≥48 mmol/mol (6.5%). Almost all children in Uganda have type 1 and need insulin from day one.
2. DKA (ISPAD 2022): glucose >11 mmol/L + venous pH <7.3 or bicarbonate <18 mmol/L + ketones (blood BOHB ≥3 mmol/L or urine ketones ++). Severity: mild pH <7.3 (HCO_3 <18), moderate pH <7.2 (HCO_3 <10), severe pH <7.1 (HCO_3 <5).
3. DKA fluids: shock → 0.9% saline 20 mL/kg rapidly; no shock → 10–20 mL/kg over 20–60 min (NICE: 10 mL/kg over 1 h). Then deficit (5% mild–moderate, 7–10% severe) plus maintenance evenly over 24–48 h (NICE: 48 h).
4. DKA insulin: soluble insulin 0.05–0.1 units/kg/h IV, started 1–2 hours AFTER fluids begin; NO insulin bolus. UCG 2023: 0.05 units/kg/h under 5 years, 0.1 units/kg/h over 5; hourly IM/SC soluble insulin 0.1 unit/kg where there is no infusion pump.
5. Potassium: add 40 mmol/L to fluids once K is known <5.5 mmol/L and the child passes urine (UCG: 20 mmol/L if K 3.5–5.5; 40 mmol/L if <3.5). Add 5% dextrose when glucose falls to about 14–17 mmol/L; aim for a fall of 2–5 mmol/L per hour.
6. Cerebral oedema (headache, falling GCS, bradycardia + hypertension, irritability) is the main cause of DKA death in children: head up, reduce fluids, mannitol 0.5–1 g/kg over 10–15 min or 3% saline 2.5–5 mL/kg; do not wait for a scan.
7. Insulin in type 1 diabetes: total daily dose about 0.5 units/kg/day at diagnosis (UCG: start 0.5 under 5 years), 0.7–1 prepubertal, 1–1.5 in puberty; premixed 30/70 twice daily (2/3 morning, 1/3 evening) is common in Uganda, basal-bolus is preferred where supported. HbA1c target <7% (53 mmol/mol).
8. Congenital hypothyroidism: about 1 in 2,000–4,000 births; prolonged jaundice, large tongue, umbilical hernia, hypotonia, constipation, large posterior fontanelle, hoarse cry. Levothyroxine 10–15 µg/kg/day as soon as possible (ideally within 2 weeks of birth) protects IQ.
9. Short stature: height <–2 SD (below the 3rd centile) or crossing centiles downward; mid-parental height = (father + mother)/2 + 6.5 cm for boys or – 6.5 cm for girls. Growth velocity matters more than a single height.
10. Ambiguous genitalia: do NOT assign sex at birth; examine for gonads; check glucose and electrolytes daily; commonest cause in a 46,XX baby is 21-hydroxylase CAH. Salt-wasting crisis at 1–3 weeks: vomiting, shock, Na low, K high, glucose low → 0.9% saline 20 mL/kg, dextrose, hydrocortisone IV.
11. Hydrocortisone: emergency 50–100 mg/m² IV (infant 25 mg, 1–5 years 50 mg, >5 years 100 mg), then 50–100 mg/m²/day 6-hourly; CAH maintenance 10–15 mg/m²/day in 3 doses + fludrocortisone 0.05–0.2 mg/day + salt 1–2 g/day in infants; double or triple doses when ill.
12. Nutritional rickets: bowed legs, wide wrists, rachitic rosary, delayed walking/teething, hypocalcaemic seizures; high ALP, low phosphate; 25-OH vitamin D <30 nmol/L = deficiency. Treat with vitamin D 2,000 IU/day (under 1 year) or 3,000–6,000 IU/day (1–12 years) for at least 12 weeks plus calcium 500 mg/day; calcium-deficiency rickets is common in Africa.

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In a nutshell

Endocrine disorders are less common than infections and malnutrition on a Ugandan paediatric ward, but they are **high-stakes**: they are easy to miss, they kill quickly when they present as emergencies (**DKA, adrenal crisis**), and they cause permanent disability when diagnosed late (**congenital hypothyroidism**, untreated **rickets**). Most are very treatable with cheap medicines (insulin, levothyroxine, hydrocortisone, vitamin D, calcium).

Five conditions matter most for the junior clerk:

1. **Type 1 diabetes mellitus (T1DM)**, often first diagnosed when the child arrives in **diabetic ketoacidosis (DKA)**, frequently mistaken for malaria, pneumonia or gastroenteritis.
2. **Congenital hypothyroidism (CH)**, the commonest preventable cause of intellectual disability; Uganda has **no routine newborn screening**, so diagnosis depends on clinical suspicion.
3. **Short stature**, mostly from chronic undernutrition (stunting), but sometimes the first clue to hypothyroidism, growth hormone deficiency, Turner syndrome, chronic disease or a skeletal dysplasia.
4. **Ambiguous genitalia / disorders of sex development (DSD)**, most often **congenital adrenal hyperplasia (CAH)**, which can cause a fatal **salt-wasting crisis** in the first weeks of life.
5. **Rickets**, from vitamin D and/or calcium deficiency.

> CLINICAL PEARL
> The one thing never to forget: **a child with deep sighing breathing, vomiting, dehydration and drowsiness may have DKA, not malaria or pneumonia. Dip the urine and check the glucose.** And in DKA, **fluids first, insulin an hour later, no insulin bolus, and watch for cerebral oedema.**

Definitions & classification

Diabetes

Term	Definition
Diabetes mellitus (WHO/ISPAD)	Symptoms plus **random plasma glucose ≥11.1 mmol/L**; or **fasting ≥7.0 mmol/L**; or **2-h OGTT ≥11.1 mmol/L**; or **HbA1c ≥48 mmol/mol (6.5%)**. Without symptoms, confirm on a second day
Type 1 diabetes	Autoimmune destruction of beta cells → absolute insulin deficiency; >90% of childhood diabetes; needs insulin for life
Type 2 diabetes	Insulin resistance; rising in obese adolescents in urban Uganda; acanthosis nigricans
Other types	Monogenic diabetes (MODY, neonatal diabetes <6 months), cystic fibrosis, drug-induced (steroids), malnutrition-related diabetes
DKA (ISPAD 2022)	**Hyperglycaemia >11 mmol/L + venous pH <7.3 or bicarbonate <18 mmol/L + ketonaemia (BOHB ≥3 mmol/L) or moderate/large ketonuria**
DKA severity	**Mild** pH <7.3 or HCO_3 <18; **moderate** pH <7.2 or HCO_3 <10; **severe** pH <7.1 or HCO_3 <5 mmol/L
HHS (hyperglycaemic hyperosmolar state)	Glucose >33.3 mmol/L, osmolality >320 mOsm/kg, little ketosis, pH >7.25; rare in children (obese adolescents with T2DM)
Hypoglycaemia in diabetes (ISPAD 2022)	Level 1 <3.9 mmol/L; level 2 <3.0 mmol/L; level 3 severe (needs help)

> **NOTE**
> NICE NG18 (2020/2023) uses slightly different bicarbonate cut-offs (mild <15, moderate <10, severe <5 mmol/L) with the same pH cut-offs. ISPAD 2022 raised the diagnostic bicarbonate threshold to **<18 mmol/L**. Quote ISPAD 2022 and note the difference.

Thyroid

Term	Definition
Congenital hypothyroidism	Thyroid hormone deficiency present at birth: thyroid **dysgenesis** (absent, small or ectopic gland, ~85%), **dyshormonogenesis** (~10–15%), or central (pituitary)
Endemic cretinism	Severe intellectual disability, deafness, spasticity and short stature from maternal and fetal **iodine deficiency** (historically in parts of western and south-western Uganda)
Acquired hypothyroidism	Usually autoimmune (Hashimoto's) in older children; more common in **Down syndrome**, Turner syndrome and type 1 diabetes
Hyperthyroidism	Usually Graves' disease in adolescents (goitre, weight loss, tremor, tachycardia, exophthalmos)

Growth

Term	Definition
Short stature	Height (length) **below −2 SD** (Z-score <−2, roughly <3rd centile) for age and sex, or **height well below the mid-parental target**, or **falling across centiles**
Stunting (WHO)	Height-for-age Z-score **<−2**; **severe stunting <−3**. Reflects chronic undernutrition and repeated infection
Mid-parental height (MPH)	(Father's height + mother's height) ÷ 2, **+6.5 cm for boys, −6.5 cm for girls**; target range ±8.5 cm (about ±2 SD)
Growth velocity	cm per year, measured over at least 6 (ideally 12) months; a falling velocity is more worrying than a short but steadily growing child
Bone age	Skeletal maturity from an X-ray of the left hand and wrist; delayed in constitutional delay, hypothyroidism, GH deficiency, malnutrition; normal in familial short stature
Proportionate vs disproportionate	Disproportionate (short limbs or short trunk) suggests a skeletal dysplasia (e.g. achondroplasia) or rickets

Sex development

Term	Definition
Ambiguous genitalia	External genitalia that do not look clearly male or female, e.g. clitoromegaly with fused labia, micropenis with hypospadias and undescended testes, bifid scrotum
DSD (disorder of sex development)	Congenital condition where chromosomal, gonadal or anatomical sex is atypical. Classified as **46,XX DSD** (commonest: CAH), **46,XY DSD** (undervirilisation: androgen insensitivity, 5-alpha-reductase deficiency, testicular dysgenesis), **sex chromosome DSD** (Turner 45,X; Klinefelter 47,XXY; mixed gonadal dysgenesis 45,X/46,XY)
CAH	Autosomal recessive defect in cortisol synthesis; **21-hydroxylase deficiency causes about 90–95%**; low cortisol (± aldosterone) and excess adrenal androgens
Salt-wasting CAH	About three-quarters of classic 21-hydroxylase deficiency; aldosterone deficiency → **hyponatraemia, hyperkalaemia, dehydration, shock** at **1–3 weeks** of age

Rickets

Term	Definition
Rickets	Failure of mineralisation of the **growth plate** in a growing child (osteomalacia is the same process in bone matrix and can occur at any age)
Nutritional rickets	From **vitamin D deficiency**, **dietary calcium deficiency**, or both (Global Consensus 2016)
Vitamin D status (25-OH-D)	**Deficient <30 nmol/L** (12 ng/mL); insufficient 30–50 nmol/L; sufficient >50 nmol/L (20 ng/mL)
Other causes	X-linked hypophosphataemic rickets (normal calcium, very low phosphate), renal rickets (CKD), vitamin D–dependent rickets, drug-induced (phenytoin, phenobarbital)

Epidemiology & Uganda context

- **Type 1 diabetes**: global incidence is rising; sub-Saharan Africa has many undiagnosed children who **die of DKA before diagnosis** (misdiagnosed as malaria, sepsis or pneumonia). In Ugandan hospitals, **most new cases present in DKA**, often severe.
- Insulin, glucometers and strips are supplied to registered children through national and partner programmes (for example **Changing Diabetes in Children** and **Life for a Child**) at many regional and national referral hospitals; supply interruptions, cost of food, distance to clinic and school stigma remain big problems.
- **Congenital hypothyroidism**: occurs in about 1 in 2,000–4,000 births worldwide. Uganda does **not have national newborn TSH screening**, so babies are diagnosed late (months), after brain damage. Uganda's **universal salt iodisation** (since the 1990s) has largely controlled iodine-deficiency goitre and cretinism, though goitre is still seen in some highland districts.
- **Short stature**: **stunting** affects about a quarter of Ugandan children under 5 (UDHS 2022), with higher rates in the south-west and Karamoja. Most short children are short because of **chronic undernutrition, recurrent infection, HIV, TB, sickle cell disease or chronic kidney or heart disease**. Endocrine causes are uncommon but treatable.
- **CAH and DSD**: rare but important; families face strong social and cultural pressures about the sex of the baby. Karyotyping and 17-hydroxyprogesterone are available mainly at Mulago/private labs (often sent abroad), so diagnosis is delayed.
- **Rickets**: still seen in Uganda despite abundant sunshine, especially in infants kept indoors and wrapped up, exclusively breastfed infants of vitamin D–deficient mothers beyond 6 months without enough complementary foods, children with **low dietary calcium** (little milk), dark skin, malnutrition, and those on long-term anticonvulsants or with chronic liver or kidney disease.

UGANDA CONTEXT

DKA in Uganda: **infusion pumps, blood gases and ketone meters are scarce** outside Mulago and a few RRHs; many units manage DKA with **0.9% saline by burette, hourly glucometer readings, urine dipstick ketones and hourly IM/SC soluble insulin**. UCG 2023 (section 8.1.4) gives children **0.9% saline 10–20 mL/kg** first, then 5–15 mL/kg/h according to response, **soluble insulin 0.05 units/kg/h (<5 years) or 0.1 units/kg/h (>5 years)**, switch to **5% dextrose when glucose <14 mmol/L**, and **potassium chloride 20 mmol/L (1 ampoule) once urine is passed** if K cannot be measured. ISPAD 2022 is more detailed (deficit over 24–48 h, potassium 40 mmol/L). Know both, and follow your ward protocol (Mulago and Mengo have paediatric DKA charts).

Pathophysiology

Type 1 diabetes and DKA

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Autoimmune β-cell destruction → absolute INSULIN DEFICIENCY
          + counter-regulatory hormones (glucagon, cortisol, catecholamines, GH) ↑
                  │                                       │
   Glucose can't enter cells;                  Lipolysis ↑ → free fatty acids
   liver makes more (gluconeogenesis)          → liver makes KETONES (BOHB, acetoacetate)
                  │                                       │
   HYPERGLYCAEMIA → osmotic diuresis           KETOACIDOSIS (pH ↓, HCO₃ ↓)
   → polyuria, thirst, DEHYDRATION,            → Kussmaul breathing, vomiting,
     loss of Na, K, PO₄                          abdominal pain, acetone breath
                  └──────────► shock, AKI, coma ◄──────┘
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- **Total-body potassium is always depleted** (urine losses, vomiting), even if the serum K is normal or high at presentation (acidosis shifts K out of cells). **Insulin and correcting acidosis drive K back into cells**, so serum K falls sharply during treatment: hence potassium in the fluids from early on.
- **Sodium**: measured Na is falsely low because glucose draws water into the blood. **Corrected Na = measured Na + 0.3 × (glucose – 5.5) mmol/L** (ISPAD uses 2 mmol/L for each 5.5 above 5.5). A corrected Na that **falls** during treatment is a warning sign for cerebral oedema.
- **Cerebral oedema** occurs in about **0.5–1%** of paediatric DKA episodes and causes most DKA deaths in children. Risk factors: young age, new diagnosis, severe acidosis, high urea, low pCO_2, falling corrected sodium, bicarbonate treatment, and (historically) large fluid volumes and insulin given early. The **PECARN FLUID trial** (NEJM 2018) found that **neither the rate nor the sodium content** of fluids (within the ranges tested) changed neurological outcome, so the focus is on careful, steady correction, not fluid starvation.

Congenital hypothyroidism

Thyroid hormone is essential for **brain myelination and neuronal growth in the first 2–3 years**. Maternal T4 crossing the placenta protects the fetus partly, so babies look **normal at birth** and signs appear over weeks. Each month of delay in treatment costs IQ points; treatment started in the **first 2 weeks** gives near-normal intelligence.

CAH (21-hydroxylase deficiency)

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Cholesterol → … → 17-OH-progesterone ──✖ 21-hydroxylase ──→ 11-deoxycortisol → CORTISOL ↓
                  progesterone ─────✖ 21-hydroxylase ──→ … → ALDOSTERONE ↓ (salt-wasting)
Low cortisol → ACTH ↑ → adrenal hyperplasia → precursors diverted → ANDROGENS ↑
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- **46,XX girls**: virilised genitalia at birth (clitoromegaly, fused labia, common urogenital sinus) with **normal internal female organs** (uterus, ovaries).
- **46,XY boys**: normal genitalia (perhaps hyperpigmented scrotum), so the first sign is a **salt-wasting crisis at 1–3 weeks**: vomiting, poor feeding, weight loss, dehydration, shock, **low Na, high K, low glucose**. Many die undiagnosed ("sepsis").
- Later: rapid growth and advanced bone age in childhood, early pubic hair, then **short adult height** from early epiphyseal fusion.

Rickets

Vitamin D (from skin exposure to UVB and from diet) is converted to 25-OH-D in the liver and $1,25\text{-}(OH)_2D$ in the kidney, which increases gut **calcium and phosphate absorption**. Deficiency → low calcium → **secondary hyperparathyroidism** → phosphate wasting in urine → **low phosphate** → the growth plate cannot mineralise →

widened, cupped, frayed metaphyses and soft, bending bones. Low dietary calcium does the same even with adequate vitamin D.

Clinical acumen

Focused history

Question	Why it matters
Polyuria, polydipsia, nocturia, new bed-wetting, weight loss, tiredness	Classic T1DM; ask specifically, since parents rarely volunteer them
Vomiting, abdominal pain, fast/deep breathing, drowsiness	DKA (often misdiagnosed as gastroenteritis, malaria, pneumonia, appendicitis)
Recent infection, missed insulin doses, insulin supply	DKA triggers in known diabetics
Newborn: prolonged jaundice, poor feeding, constipation, sleepy, hoarse cry	Congenital hypothyroidism
Vomiting, poor weight gain, collapse at 1–3 weeks	Salt-wasting CAH
Previous neonatal deaths, consanguinity	Autosomal recessive conditions (CAH, dyshormonogenesis)
Birth weight, gestation, growth chart (child health card)	Short stature: SGA, timing of growth failure
Parents' heights, age of parents' puberty	Familial short stature, constitutional delay
Diet (milk, eggs, fish, complementary foods), sunlight exposure, breastfeeding duration	Rickets, stunting
Chronic illness: HIV, TB, sickle cell, heart/kidney disease, chronic diarrhoea, steroids	Secondary short stature, poor growth
Development, school performance, headaches, vision	Hypothyroidism, intracranial tumour (craniopharyngioma) with GH deficiency
Puberty: age of breast development/menarche or testicular enlargement	Precocious puberty (girls <8, boys <9) or delayed puberty (girls >13, boys >14)
Immunisation, HIV status, feeding	Background in every child

Examination

- **Anthropometry**: weight, height/length (use a length board under 2 years), **plot on WHO charts**, Z-scores (WAZ, HAZ, WHZ/BMI), MUAC, head circumference; **arm span** and **upper:lower segment ratio** for proportion; **parents' heights**.
- **DKA**: hydration and shock signs (pulse, CRT, BP), **Kussmaul breathing** (deep, sighing), **acetone smell**, GCS, signs of infection; weigh (or use a recent weight).
- **Hypothyroidism (infant)**: coarse facies, **large tongue**, **umbilical hernia**, **large posterior fontanelle** (>0.5 cm), hypotonia, dry skin, mottling, prolonged jaundice, goitre (dyshormonogenesis), hypothermia, bradycardia.
- **Short stature clues**: dysmorphism (Turner: webbed neck, wide carrying angle, widely spaced nipples, lymphoedema; Down syndrome), midline defects (hypopituitarism), goitre, Cushingoid features (steroids), signs of chronic disease (clubbing, pallor, hepatosplenomegaly), rickets, disproportion.
- **Genitalia**: phallus length (stretched penile length <2.5 cm at term = micropenis), position of the urethral opening, labioscrotal fusion, **palpable gonads** (a palpable gonad is almost always a testis, so 46,XY or mosaic), pigmentation; **Prader stage** of virilisation. Check BP and hydration.
- **Rickets**: frontal bossing, delayed fontanelle closure, **craniotabes** (ping-pong skull), delayed dentition, **rachitic rosary** (swollen costochondral junctions), **Harrison's sulcus**, **wrist widening**, bowed legs (genu

varum) or knock knees (genu valgum), hypotonia, delayed walking, tetany.

Red flags

RED FLAG

- **DKA with GCS falling, headache, irritability, bradycardia with rising BP, abnormal posturing, incontinence, or cranial nerve palsy** = cerebral oedema; treat now.
- DKA in a child **under 5 years**, **pH <7.1**, shock, or corrected Na falling during treatment.
- **Infant aged 1–3 weeks with vomiting, shock, Na <130 and K >6** (± ambiguous genitalia or a pigmented scrotum) = adrenal crisis.
- **Bilateral impalpable testes in an apparently male newborn** = possible virilised 46,XX CAH girl; do not circumcise or register sex yet.
- **Short child with falling growth velocity, headache or visual field loss** (craniopharyngioma), or with **weight gain** (hypothyroidism, Cushing's).
- **Hypocalcaemic seizures** or **heart failure** (dilated cardiomyopathy) in an infant with rickets.

Thinking like a clinician

- **"Tired, thin, thirsty, toilet"**: the 4 Ts of new T1DM. A child with **acidotic breathing and a clear chest** has acidosis (DKA, severe malaria, dehydration, AKI, salicylate), not pneumonia.
- **The glucose meter in a Ugandan emergency room is usually used for low glucose; use it for high glucose too.** "HI" on a glucometer means >33 mmol/L.
- **In DKA, the danger is not the glucose but the fluid, potassium and brain.** Glucose falls quickly with fluids alone; acidosis resolves more slowly; insulin is needed to switch off ketone production.
- **A short child growing at a normal velocity, with a short family and normal bone age** has familial short stature. **A short child with delayed bone age, normal velocity and a parent who was a "late bloomer"** has constitutional delay. **A child whose velocity falls** has a disease until proven otherwise.
- **In Uganda, the commonest causes of short stature are malnutrition and chronic disease**: check HIV, TB, sickle cell, renal function and diet before thinking of growth hormone.
- **Ambiguous genitalia is a medical emergency and a social one**: check glucose and electrolytes, avoid guessing the sex in front of the family, use neutral words ("your baby"), and refer to the paediatric endocrine/surgical team.
- **Common student mistakes**: insulin bolus in DKA; forgetting potassium; stopping the insulin infusion when glucose normalises (instead add dextrose); measuring short stature without plotting on a chart or ignoring parents' heights; telling parents "it's a boy" in a baby with impalpable testes; giving vitamin D without calcium in a child with poor diet.

EXAM FAVOURITE

Viva favourite: **"Why don't we give an insulin bolus in DKA, and why wait an hour?"** A bolus does not speed resolution of acidosis, causes a rapid fall in glucose and osmolality (cerebral oedema risk) and drops potassium suddenly. Starting insulin **1–2 hours after fluids** (ISPAD/NICE) was associated with less cerebral oedema in UK data, and glucose falls substantially with fluids alone.

Clinical workup

Step 0: First 5 minutes (suspected DKA or adrenal crisis)

1. **ABC (ETAT)**: airway (vomiting, reduced GCS: recovery position, NG tube if vomiting and drowsy), oxygen if SpO_2 <94% or shocked.
2. **Circulation**: shock (cold, CRT >3 s, weak fast pulse, or hypotension) → **0.9% saline 20 mL/kg** rapidly (DKA: ISPAD; NICE: 10 mL/kg, repeat if needed; adrenal crisis 20 mL/kg).
3. **Glucose** (glucometer) and **urine/blood ketones**; **weigh** the child.
4. Bloods **before treatment**: venous gas (pH, HCO_3, pCO_2) if available, electrolytes (Na, K), urea/creatinine, FBC, glucose (lab), malaria test, blood culture if febrile; for adrenal crisis also save serum for **cortisol and 17-OHP** before hydrocortisone.
5. **GCS** and neurological observations (baseline for cerebral oedema).
6. **Adrenal crisis suspected** (shock + low Na + high K ± hypoglycaemia in an infant) → **hydrocortisone IV now** (infant 25 mg).
7. Call senior; start the DKA flow chart and an hourly observation sheet.

> **RED FLAG**
> Call the senior and anaesthetist immediately for: GCS ≤12 or falling, shock, pH <7.1, age under 5 with DKA, K <3 or >6 mmol/L, signs of cerebral oedema, or an infant in adrenal crisis.

Step 1: Focused history

- [ ] **Duration of polyuria, polydipsia, weight loss** → new T1DM; length of illness correlates with severity.
- [ ] **Known diabetic: insulin type, dose, last dose, missed doses, supply, glucose diary** → trigger and future plan.
- [ ] **Vomiting, abdominal pain, breathing, drowsiness, headache** → DKA severity, cerebral oedema.
- [ ] **Infection symptoms** (fever, cough, dysuria, diarrhoea) → precipitant; malaria test in every febrile child.
- [ ] **Neonate**: feeding, vomiting, weight, jaundice duration, stool, genital appearance, family deaths → CAH, CH.
- [ ] **Short child**: birth weight, growth records, parents' heights, puberty in parents, diet, chronic illness, medicines (steroids), development, headaches/vision.
- [ ] **Rickets**: sunlight, diet (milk, calcium), breastfeeding without complementary foods, anticonvulsants, chronic diarrhoea, liver or kidney disease.
- [ ] **Immunisation, HIV status of mother and child, feeding** → every child.

Step 2: Focused examination

General: weight, height/length, Z-scores, MUAC, temperature, hydration, pallor, jaundice, pigmentation, dysmorphism.

DKA: pulse, CRT, BP, RR and pattern (Kussmaul), acetone breath, GCS and pupils, abdomen (tenderness common), focus of infection.

Endocrine-specific: thyroid (goitre), genitalia (phallus, urethral opening, labioscrotal fusion, gonads), pubertal staging (Tanner), skeletal signs of rickets, body proportions.

Document these positives	And these negatives
Weight (used for all calculations) and degree of dehydration	"No signs of shock"
RR/pattern, GCS, pupils (baseline)	"No headache, no bradycardia or hypertension"
Height, Z-scores, MPH, growth velocity	"No dysmorphic features; proportionate"

Document these positives	And these negatives
Genitalia description (phallus length, gonads, Prader stage)	"Gonads not palpable" or "both testes in scrotum"
Rickets signs (wrists, rosary, legs)	"No craniotabes; no tetany"

Step 3: Bedside tests

- **Capillary glucose** (glucometer reads "HI" above about 33 mmol/L).
- **Urine dipstick**: glucose and **ketones** (++ or more supports DKA); blood ketone meter (BOHB ≥3 mmol/L) where available.
- **Urine output** (catheter or weigh nappies) and fluid balance chart.
- **Blood pressure** with the right cuff (CAH 11-beta-hydroxylase deficiency causes hypertension; cerebral oedema raises BP).
- **Hb, mRDT** (fever in DKA may be malaria or sepsis).
- **ECG** if available in DKA (peaked T waves = hyperkalaemia; flat T, U waves = hypokalaemia).

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management	Uganda availability
Venous blood gas	DKA	pH <7.3, HCO_3 <18, low pCO_2	Severity → deficit 5% vs 10%; monitoring 2–4 hourly	Mulago, some RRHs/private; otherwise use clinical severity and bicarbonate on electrolytes
Electrolytes, urea, creatinine	DKA, adrenal crisis, vomiting infant	K high/low; Na low (correct for glucose); urea high	Potassium in fluids; corrected Na trend; AKI	RRH/national referral
HbA1c	New diabetes, follow-up every 3 months	≥48 mmol/mol (6.5%) supports chronic hyperglycaemia	Control assessment	Many hospitals and private labs
TSH, free T4	Suspected hypothyroidism, short stature, Down syndrome (annually)	High TSH + low FT4 = primary hypothyroidism	Levothyroxine	RRH/private; often sent out
Karyotype	Ambiguous genitalia, short girl (Turner)	46,XX, 46,XY, 45,X, mosaics	Diagnosis, sex assignment	Mulago/private, send-out; weeks
17-OH-progesterone (≥24–48 h of age)	Ambiguous genitalia, salt-losing infant	Markedly raised in 21-OHD	Confirms CAH	Send-out lab
Pelvic/abdominal ultrasound	Ambiguous genitalia	Uterus present (46,XX CAH) or absent; gonads	Diagnosis	Widely available
Bone age (left hand/wrist X-ray)	Short stature, puberty disorders	Delayed (CDGP, hypothyroid, GHD) or normal (familial)	Prognosis, cause	Hospitals
Wrist/knee X-ray	Rickets	Cupping, splaying, fraying of metaphyses; osteopenia	Confirms rickets, monitors healing	Hospitals

Test	When	Expected / abnormal finding	How it changes management	Uganda availability
Calcium, phosphate, ALP, PTH, 25-OH-D	Rickets	Ca low/normal, PO_4 low, ALP high, PTH high, 25-OHD <30 nmol/L	Vitamin D ± calcium; XLH if very low PO_4 with normal Ca and PTH	Ca/PO_4/ALP at hospitals; PTH and 25-OHD send-out
FBC, ESR, renal function, LFTs, HIV, stool, coeliac serology	Short stature screen	Chronic disease markers	Treat underlying disease	Hospitals
Thyroid ultrasound / isotope scan	CH (specialist)	Absent/ectopic gland	Prognosis, genetic counselling	Mulago/private

Step 5: Putting it together

Model one-line summary: "A 9-year-old girl, 24 kg, with 3 weeks of polyuria, polydipsia and weight loss, now vomiting with Kussmaul breathing; glucose 28 mmol/L, urine ketones +++, pH 7.08: **severe DKA at first presentation of type 1 diabetes**, no shock, GCS 14."

NOTE

Worked example: DKA fluid prescription (ISPAD 2022 / NICE 2020). 24 kg girl, **severe DKA** (pH 7.08), not shocked, received a 10 mL/kg bolus (240 mL) over 1 hour.

- **Deficit**: severe → assume **10%** (NICE; ISPAD 7–10%): 24 kg × 10% = **2,400 mL**. NICE subtracts a non-shock bolus: 2,400 − 240 = **2,160 mL**.
- **Maintenance** (Holliday–Segar): 100 × 10 + 50 × 10 + 20 × 4 = **1,580 mL/day** → 3,160 mL over 48 h.
- **Total over 48 h** = 2,160 + 3,160 = 5,320 mL → **about 110 mL/h** of 0.9% saline (with potassium 40 mmol/L once K <5.5 and urine passed).
- **Insulin**: start 1–2 h after fluids: **0.05–0.1 units/kg/h = 1.2–2.4 units/h** (50 units soluble insulin in 50 mL 0.9% saline = 1 unit/mL → 1.2–2.4 mL/h).
- When glucose reaches about **14 mmol/L**, change to **0.9% saline with 5% dextrose** (+ KCl) at the same rate; keep the insulin running.

NOTE

Worked example: short stature. A 10-year-old boy is 124 cm (HAZ −2.3). Father 160 cm, mother 152 cm → MPH = (160 + 152)/2 + 6.5 = **162.5 cm** (target range 154–171 cm). His growth record shows steady growth along the same Z-score line (5 cm/year), bone age 8 years, and the father "grew late". **Interpretation**: **constitutional delay of growth and puberty** (delayed bone age, normal velocity, family history), with some familial component; reassure and follow. If his velocity had fallen to 2 cm/year, you would screen for hypothyroidism, chronic disease, GH deficiency and an intracranial tumour.

Differential diagnosis

A child with acidotic breathing, vomiting and dehydration

Condition	Distinguishing features	Key investigation
DKA	Polyuria/polydipsia before, weight loss, glucose >11, ketonuria	Glucose, urine ketones, gas

Condition	Distinguishing features	Key investigation
Severe malaria with acidosis	Fever, parasitaemia, anaemia, hypoglycaemia more likely than hyperglycaemia	mRDT/smear
Gastroenteritis with severe dehydration	Diarrhoea predominant, glucose normal, ketones mild (starvation)	Glucose (stress hyperglycaemia is usually <11)
Pneumonia	Fever, crackles, focal signs, hypoxaemia	SpO_2, CXR
Sepsis/meningitis	Fever, neck stiffness, purpura	Cultures, LP
Salicylate poisoning	History, tinnitus, mixed acid–base	History, level
Acute abdomen (appendicitis)	Localised tenderness, guarding; DKA abdominal pain improves with treatment	Re-examine after rehydration

Short stature

Cause	Clues	Key test
Familial	Short parents, normal velocity, bone age = chronological age	MPH, bone age
Constitutional delay	Normal velocity, delayed bone age and puberty, family history of late puberty	Bone age
Nutritional (stunting)/chronic disease	Low weight-for-height first, poor diet, infections, HIV, TB, CKD, sickle cell, IBD/coeliac	Diet history, FBC, HIV, renal function
Hypothyroidism	Falling velocity, weight gain, delayed bone age, constipation, school decline	TSH, FT4
GH deficiency	Chubby, immature face, poor velocity; midline defects; neonatal hypoglycaemia; tumour symptoms	IGF-1, stimulation tests (specialist)
Turner syndrome (girls)	Dysmorphism, lymphoedema, no puberty, heart murmur (coarctation)	Karyotype
Cushing's/steroids	Obese, short, striae, hypertension, steroid creams/tablets	History, cortisol
Skeletal dysplasia	Disproportion	Skeletal survey
Rickets	Bowed legs, wide wrists	ALP, PO_4, X-ray
SGA without catch-up	Low birth weight, no catch-up by 2–4 years	Records

Ambiguous genitalia

Cause	Clues	Key test
21-hydroxylase CAH (46,XX)	Impalpable gonads, uterus on ultrasound, pigmentation, salt-wasting at 1–3 weeks	17-OHP, electrolytes
Maternal androgens (46,XX)	Maternal virilisation, androgenic drugs	History
Partial androgen insensitivity / 5-alpha-reductase deficiency (46,XY)	Palpable gonads, no uterus	Karyotype, hormones (specialist)
Mixed gonadal dysgenesis (45,X/46,XY)	Asymmetric gonads, Turner features	Karyotype
Severe hypospadias with undescended testes	Usually 46,XY; always investigate if both testes impalpable	Karyotype, ultrasound

Bowed legs

Cause	Clues	Key test
Physiological genu varum	Under 2 years, symmetrical, improving	Clinical follow-up
Nutritional rickets	Wide wrists, rosary, high ALP	X-ray, biochemistry
Blount disease	Early walkers, obese, unilateral or asymmetrical, abrupt angulation	X-ray
X-linked hypophosphataemic rickets	Family history, very low PO_4, normal Ca, poor response to vitamin D	PO_4, family screening
Skeletal dysplasia	Disproportion, short stature	Skeletal survey

Investigations

Test	What you are looking for	Why
Glucose, urine/blood ketones	DKA vs hyperglycaemia without ketosis	Diagnosis
Venous gas, electrolytes, urea	Acidosis severity, K, corrected Na	Fluids, potassium, cerebral oedema risk
Malaria test, cultures, CXR, urine	Precipitating infection	Treat trigger
HbA1c	Duration/control	Baseline and follow-up
TSH/FT4	Hypothyroidism	Levothyroxine
Karyotype, 17-OHP, ultrasound	DSD/CAH	Diagnosis and sex assignment
Bone age	Growth potential	Short stature and puberty
Ca, PO_4, ALP, PTH, 25-OHD, X-ray	Rickets type and severity	Vitamin D/calcium vs other causes

Management

A. Diabetic ketoacidosis (ISPAD 2022, NICE NG18, UCG 2023)

```
Confirm DKA: glucose >11 + pH <7.3 / HCO₃ <18 + ketones
   |
SHOCK? ── yes ──► 0.9% saline 20 mL/kg rapidly, repeat if needed (senior)
   | no
0.9% saline 10–20 mL/kg over 20–60 min (NICE: 10 mL/kg over 60 min)
   |
Deficit (mild–moderate 5%; severe 7–10%) + maintenance over 24–48 h
 (0.9% saline, or 0.45–0.9% saline per protocol) + KCl 40 mmol/L once K <5.5 and urine passed
   |
After 1–2 h of fluids: soluble insulin 0.05–0.1 units/kg/h IV (NO bolus)
   |
Glucose ≈14–17 mmol/L or falling >5 mmol/L/h → add 5% dextrose (10% if needed); keep insulin
   |
Hourly: glucose, vitals, GCS, fluid balance · 2–4 hourly: gas, Na, K
   |
Resolution: pH >7.3, HCO₃ >18, BOHB <1 (or urine ketones falling), eating
   → SC insulin, stop IV insulin 30–60 min after first SC dose
```

DKA drug and fluid table

Treatment	Dose	Route	Frequency / duration	Max / notes
0.9% saline (shock)	**20 mL/kg** (NICE: 10 mL/kg, discuss before more)	IV rapidly	Repeat if still shocked	Shock bolus not subtracted from deficit (NICE)
0.9% saline (no shock)	**10–20 mL/kg** (UCG 10–20; NICE 10 mL/kg over 60 min)	IV over 20–60 min	Once	NICE subtracts this bolus from the deficit
Rehydration	Deficit **5%** (pH ≥7.1) or **7–10%** (pH <7.1) of body weight **+ maintenance**	IV evenly over **24–48 h**	Continuous	Do not include ongoing urine losses; avoid exceeding 1.5–2 × maintenance
Soluble (regular) insulin	**0.05–0.1 units/kg/h** (UCG: 0.05 under 5 years; 0.1 over 5 years)	IV infusion (50 units in 50 mL 0.9% saline = 1 unit/mL)	Start **1–2 h after fluids**; continue until DKA resolved and SC insulin given	**No bolus**; if BOHB not falling at 6–8 h, increase to 0.1 units/kg/h or more
Insulin without a pump	**0.1 unit/kg** (0.05 in young children) soluble or rapid-acting insulin	**IM or SC hourly** (or 1–2-hourly per protocol)	Until ketosis resolves	UCG 2023 and ISPAD accept SC/IM regimens where IV infusion is unsafe; perfusion must be restored first
Potassium chloride	**40 mmol/L** of IV fluid (UCG: 20 mmol/L if K 3.5–5.5; 40 mmol/L if <3.5; none if >5.5)	In the IV fluid	Throughout	Start with the first fluids if K <3.5 (hypokalaemia); once urine passed if K normal; withhold if anuric or K >5.5. Max 0.5 mmol/kg/h. KCl 15% 10 mL ampoule = 20 mmol
Dextrose	Add **5%** dextrose (10% if glucose keeps falling)	In the IV fluid	When glucose ≈**14–17 mmol/L** (UCG: <14 mmol/L) or falls >5 mmol/L/h	Aim glucose 8–12 mmol/L; never stop insulin to treat falling glucose (add dextrose)
Mannitol 20% (cerebral oedema)	**0.5–1 g/kg** (2.5–5 mL/kg)	IV over 10–15 min	Repeat after 30 min if no response	Or **3% saline 2.5–5 mL/kg** over 10–15 min
Sodium bicarbonate	**Not routine**	—	—	Only for life-threatening hyperkalaemia or cardiovascular collapse; increases cerebral oedema risk
Antibiotics / antimalarials	As per infection	—	—	DKA can cause raised WBC; look for a real focus

DRUG DOSING

Insulin infusion for a 20 kg child at 0.1 units/kg/h = 2 units/h = 2 mL/h of a 1 unit/mL solution. Run it through a dedicated line, flush the line with 20–30 mL of the insulin solution first (insulin sticks to plastic), and **never stop it because the glucose is normal**: add dextrose instead, since ketones are still being made.

Monitoring and cerebral oedema

- **Hourly**: glucose, HR, BP, RR, GCS (neuro obs), fluid input/output, insulin rate.

- **2–4 hourly**: venous gas, Na, K, urea; calculate corrected Na.
- Watch for **cerebral oedema**: headache, recurrent vomiting, irritability or lethargy, falling GCS, **inappropriate bradycardia with rising BP**, abnormal posturing, sluggish pupils, cranial nerve palsy, incontinence. Usually 4–12 h into treatment.
- **Treatment**: raise the head of the bed 30°, **reduce IV fluids by one-third**, **mannitol 0.5–1 g/kg over 10–15 min** (or **3% saline 2.5–5 mL/kg**), call senior/ICU, intubate if needed; **CT only after treatment has started**.

After DKA resolves: starting subcutaneous insulin

Regimen	How to give	Notes
Premixed 30/70 (Mixtard) twice daily (common in Uganda)	TDD ≈ **0.5–1 unit/kg/day** (UCG: start 0.5 under 5 years); **2/3 before breakfast, 1/3 before supper**, 30 min before meals	Needs regular meals and mid-morning/afternoon snacks
Basal-bolus (preferred by ISPAD where feasible)	**40–50% of TDD as basal** (NPH at night or long-acting analogue), the rest as **soluble/rapid-acting before each meal**	More flexible; needs more injections and testing
Typical TDD	Prepubertal **0.7–1 unit/kg/day**; puberty **1–1.5**; "honeymoon" (partial remission) may be <0.5	Adjust by 10–20% every few days on glucose readings

Targets (ISPAD 2022): glucose **4–10 mmol/L** most of the time; **HbA1c <7% (53 mmol/mol)**; avoid hypoglycaemia.

Diabetes education before discharge (both caregivers, and the child if old enough): insulin storage (cool place: clay pot with wet sand works when there is no fridge), injection technique and site rotation, glucose testing, recognising and treating hypoglycaemia (sugar always available; glucagon where possible), sick-day rules (never stop insulin; check ketones; drink fluids; come early), diet (regular meals, snacks), exercise, school letter, follow-up in the diabetes clinic, peer support.

Sick-day rules (ISPAD 2022)

- **Never stop insulin** even if not eating; check glucose every 2–3 hours and ketones.
- If ketones are raised and glucose high: give **extra rapid-acting insulin 10–20% of TDD** every 2–4 h.
- Drink sugar-free fluids if glucose high; sugary fluids if glucose low.
- Come to hospital for vomiting, abdominal pain, deep breathing, drowsiness, or ketones not falling.

B. Congenital hypothyroidism

1. **Suspect** in any baby with prolonged jaundice (>14 days term), poor feeding, constipation, hypotonia, large tongue, umbilical hernia, large posterior fontanelle, or a goitre; and in all babies with **Down syndrome** (check TSH at birth or by 2 weeks, then yearly).
2. Send **TSH and free T4**; **start treatment immediately** if TSH is high and FT4 low (do not wait for scans).
3. **Levothyroxine 10–15 µg/kg/day orally once daily** (about **37.5–50 µg/day** for a term baby): crush the tablet, mix with a little breast milk or water and give by spoon; **not with soy, iron or calcium** (reduce absorption).
4. **Targets**: FT4 in the upper half of normal within 2 weeks; TSH normal within 4 weeks.
5. **Monitor**: TSH and FT4 **2 and 4 weeks** after starting, then every **1–2 months in the first 6 months**, every 3–4 months until 3 years, then every 6–12 months; **plot growth and development**.
6. **Counsel**: lifelong treatment (unless transient; reassess at 3 years), daily dosing, and the excellent outlook if treated early.

> **UGANDA CONTEXT**
> Without newborn screening, Ugandan babies with CH are typically diagnosed at **several months** with established developmental delay. Your clinical suspicion is the screening test: **"prolonged jaundice + constipation + umbilical hernia + large tongue" = check TSH.** Levothyroxine 50 µg and 100 µg tablets are on the essential medicines list; halve or crush as needed.

C. Short stature

1. **Measure accurately** (length board/stadiometer), plot on **WHO charts**, calculate Z-scores, measure the parents.
2. Calculate **MPH** and compare.
3. **Assess growth velocity** over 6–12 months (bring child health card records).
4. **Look for chronic disease and malnutrition first** (diet, HIV, TB, sickle cell, renal, cardiac, gut).
5. **Screen**: FBC, renal function, LFTs, TSH/FT4, HIV, stool, urine, **bone age**; **karyotype in every short girl** (Turner syndrome can present only with short stature).
6. **Treat the cause**: nutrition rehabilitation, treat infections, levothyroxine for hypothyroidism; growth hormone for GH deficiency or Turner syndrome is specialist-only and largely unavailable/unaffordable in Uganda.
7. **Familial or constitutional delay**: reassure, monitor every 6 months; low-dose testosterone for boys with severe distress from delayed puberty (specialist).

D. Ambiguous genitalia and CAH

Immediate approach to a newborn with ambiguous genitalia

1. **Do not assign sex** or register the baby until assessed; use neutral words ("your baby"); explain this is a recognised condition and that tests will help decide; **avoid circumcision**.
2. **Examine** carefully: phallus length, urethral opening, labioscrotal fusion, **gonads** (palpable?), pigmentation, dysmorphism; BP.
3. **Investigate**: **glucose and electrolytes daily** (salt-wasting appears from day 4–5 onwards, usually by 1–3 weeks), **pelvic ultrasound** (uterus?), **karyotype**, **17-OHP** (after 24–48 h), cortisol.
4. **Watch for salt-wasting**: weight daily, feeding, vomiting.
5. **Refer** early to a paediatric endocrinologist and surgeon (Mulago); involve the family in a multidisciplinary decision about sex of rearing; psychological support.

Salt-wasting adrenal crisis

Treatment	Dose	Route	Frequency	Notes
0.9% saline	**20 mL/kg** bolus (repeat as needed), then rehydration with dextrose-saline	IV	Over the first hour, then 24–48 h	Correct hypovolaemia and hyponatraemia (raise Na no faster than about 10 mmol/L/24 h)
10% dextrose	**2–5 mL/kg** if hypoglycaemic (neonate 2 mL/kg)	IV	Then dextrose-containing fluids	Check glucose hourly
Hydrocortisone	**50–100 mg/m²** IV bolus (practical: infant **25 mg**, 1–5 years **50 mg**, >5 years **100 mg**)	IV (IM if no IV)	Then **50–100 mg/m²/day** divided **6-hourly**	Take blood for cortisol/17-OHP first if possible, but never delay

Treatment	Dose	Route	Frequency	Notes
Hyperkalaemia	Usually corrects with saline and hydrocortisone; if K >7 or ECG changes: calcium gluconate 10% 0.5 mL/kg IV over 5–10 min, salbutamol, insulin-dextrose	IV	—	ECG monitoring

CAH maintenance

Drug	Dose	Route	Frequency	Notes
Hydrocortisone	**10–15 mg/m²/day**	Oral	Divided **3 times daily**	Prednisolone/dexamethasone avoided in growing children (growth suppression)
Fludrocortisone	**0.05–0.2 mg/day**	Oral	Once or twice daily	Salt-wasting forms; monitor BP, renin
Sodium chloride	**1–2 g/day (17–34 mmol)** in infancy (Medscape: up to 2–4 g/day)	Oral, in feeds	Divided	Infants up to about 6–12 months
Sick-day ("stress") dosing	**Double or triple** the oral hydrocortisone dose for fever/illness; **IM hydrocortisone** (as above) if vomiting or collapsed	Oral/IM	Until well	Give the family an emergency letter and IM hydrocortisone if possible

E. Rickets (Global Consensus 2016)

1. **Confirm**: clinical signs + **high ALP**, low phosphate, low/normal calcium, high PTH ± **25-OHD <30 nmol/L**; wrist/knee X-ray.
2. **Treat hypocalcaemic seizures/tetany first**: **10% calcium gluconate 0.5 mL/kg (max 20 mL) IV over 5–10 minutes** with cardiac monitoring, then oral calcium.
3. **Vitamin D (cholecalciferol)** for at least **12 weeks**:

Age	Daily dose for 12 weeks	Single large dose ("stoss") alternative	Maintenance after
<3 months	**2,000 IU**	Not recommended	400 IU/day
3–12 months	**2,000 IU**	**50,000 IU** once	400 IU/day
1–12 years	**3,000–6,000 IU**	**150,000 IU** once	600 IU/day
>12 years	**6,000 IU**	**300,000 IU** once	600 IU/day

4. **Calcium 500 mg/day** (elemental, oral) **with** vitamin D in all cases (dietary calcium deficiency is common in Africa); improve dietary calcium (milk, small fish such as mukene/silver fish, eggs, green leaves).
5. **Monitor**: ALP and X-ray at **3 months** (healing: a dense provisional zone of calcification); bowing improves over 1–2 years; osteotomy only for severe persistent deformity after healing.
6. **Prevent**: sunlight exposure (face and arms outdoors for 15–30 min most days), **vitamin D 400 IU/day for infants** where deficiency is likely, adequate complementary foods with calcium.

F. Other endocrine problems to recognise

Condition	Key features	First steps
Graves' disease	Adolescent girl, goitre, weight loss, tremor, tachycardia, eye signs	TSH/FT4; carbimazole 0.25–1 mg/kg/day (max 30–40 mg) + propranolol 0.5–1 mg/kg/day in 3 doses; refer

Condition	Key features	First steps
Endemic/iodine-deficiency goitre	Diffuse goitre, euthyroid or hypothyroid	Iodised salt; TSH; refer large goitres
Precocious puberty	Breasts <8 years or testes enlargement <9 years	Bone age, LH/FSH; refer (brain MRI in boys and young girls)
Delayed puberty	No breasts by 13 or testicular enlargement by 14	Chronic disease, malnutrition, constitutional delay, Turner/Klinefelter; karyotype
Adrenal insufficiency from stopping steroids	Child on long-term prednisolone (nephrotic syndrome, asthma) becomes ill	Stress-dose hydrocortisone; taper steroids slowly
Diabetes insipidus	Polyuria with normal glucose, dilute urine, hypernatraemia	Refer (head injury, tumour, Langerhans cell histiocytosis)

Complications

Condition	Complications
DKA	**Cerebral oedema**, hypokalaemia (arrhythmias), hypoglycaemia (during treatment), aspiration, AKI, venous thrombosis, death
Type 1 diabetes (long term)	Hypoglycaemia, recurrent DKA, poor growth, delayed puberty, retinopathy, nephropathy (microalbuminuria), neuropathy, hypertension, thyroid and coeliac disease, school and psychological problems
Congenital hypothyroidism (untreated)	Intellectual disability, short stature, deafness (if iodine deficient), delayed development
CAH	Salt-wasting death, hypoglycaemia, wrong sex assignment, short adult height, precocious pseudopuberty, infertility, iatrogenic Cushing's from over-treatment
Rickets	Hypocalcaemic seizures, dilated cardiomyopathy, laryngospasm, deformities (genu varum/valgum), pelvic deformity (obstructed labour later in girls), fractures, delayed motor milestones, pneumonia (soft chest wall)

Ward-round & exam pearls

Model case presentation

> "Miss A.N., a **9-year-old girl** from Mukono, **weight 24 kg** (weight-for-age Z –1.8, BMI-for-age Z –1.5; she has lost about 3 kg in a month per her mother), height 128 cm (HAZ –0.7), presented with **3 weeks of polyuria, polydipsia, nocturia and new bed-wetting**, and **2 days of vomiting and abdominal pain**, then **fast deep breathing and drowsiness** today. She was treated for malaria at a clinic 3 days ago. On arrival: GCS 13, Kussmaul breathing, RR 36, HR 132, CRT 3 s, BP 98/60, **acetone smell**, ~10% dehydrated. **Glucose 'HI'** (lab 31 mmol/L), **urine ketones +++**, venous **pH 7.06, HCO_3 5 mmol/L**, Na 131, K 5.2, urea 9. mRDT negative. **Fully immunised** (UNEPI including HPV dose 1 at school), **eats family food**, **HIV-negative**. **Assessment: severe DKA at first presentation of type 1 diabetes. Plan**: 0.9% saline 10 mL/kg (240 mL) over 1 hour then deficit (10%) plus maintenance over 48 h at 110 mL/h with KCl 40 mmol/L once urine passed; soluble insulin 0.1 units/kg/h (2.4 units/h) starting 1 hour after fluids, no bolus; add 5% dextrose at glucose 14 mmol/L; hourly glucose, GCS and vitals, 2-hourly gas and electrolytes; watch for cerebral oedema; diabetes education and SC insulin when DKA resolves."

Viva questions with answers

1. **Define DKA (ISPAD 2022).** Glucose >11 mmol/L, venous pH <7.3 or HCO_3 <18 mmol/L, and ketonaemia (BOHB ≥3) or moderate–large ketonuria.
2. **Why is potassium added even when serum K is normal?** Total-body K is depleted; insulin and correction of acidosis shift K into cells, causing dangerous hypokalaemia.
3. **What are the signs of cerebral oedema and its treatment?** Headache, irritability, falling GCS, bradycardia with hypertension, abnormal posture; mannitol 0.5–1 g/kg or 3% saline 2.5–5 mL/kg, head up, reduce fluids, ICU.
4. **When do you add dextrose to the DKA fluids?** When glucose falls to about 14–17 mmol/L (UCG <14), keeping the insulin running.
5. **Dose of levothyroxine in congenital hypothyroidism?** 10–15 µg/kg/day, started as soon as the diagnosis is made.
6. **Why must a short girl have a karyotype?** Turner syndrome may present only with short stature, and growth hormone and puberty induction (plus cardiac and renal screening) change the outlook.
7. **Newborn with clitoromegaly, fused labia, no palpable gonads: what is the diagnosis and the danger?** 46,XX CAH (21-hydroxylase deficiency) until proven otherwise; salt-wasting crisis at 1–3 weeks.
8. **Treatment of nutritional rickets?** Vitamin D 2,000–6,000 IU/day (by age) for at least 12 weeks or a stoss dose, with calcium 500 mg/day, then maintenance.

OSCE tips

- **Growth station**: plot correctly (age in completed months; length lying under 2 years), state Z-scores, calculate MPH, comment on velocity and proportion.
- **DKA station**: say the sequence "**ABC, weigh, glucose and ketones, gas and electrolytes, fluids, insulin after 1–2 hours, potassium, dextrose at 14, hourly obs, cerebral oedema**".
- **Counselling a newly diagnosed diabetic family**: insulin storage, injection, hypoglycaemia, sick-day rules, school, follow-up; check understanding.

Mnemonics & memory aids

- **New T1DM: the 4 Ts**: **T**oilet (polyuria), **T**hirsty, **T**ired, **T**hinner.
- **DKA order: "Fluids → (1 h) → Insulin → K → Dextrose at 14 → Watch the brain"**.
- **DKA diagnostic triad "D-K-A"**: **D**iabetes (glucose >11), **K**etones, **A**cidosis (pH <7.3/HCO_3 <18).
- **Severity by pH "7.3-7.2-7.1"**: mild, moderate, severe.
- **Congenital hypothyroidism "the 6 Ps"**: **P**rolonged jaundice, **P**oor feeding, **P**ot belly with umbilical hernia, **P**rotruding tongue, **P**uffy face, **P**osterior fontanelle large (plus constipation and hypotonia).
- **CAH crisis "Low Na, high K, low glucose, low BP"** at **1–3 weeks**.
- **MPH**: "(Dad + Mum) ÷ 2, **+6.5 boys, –6.5 girls**".
- **Rickets signs head to toe**: bossing and craniotabes → delayed teeth → rosary and Harrison's sulcus → wide wrists → bowed legs.

High-yield summary

Numbers & doses to know

Item	Value
Diabetes diagnosis	Random glucose **≥11.1 mmol/L** + symptoms; fasting **≥7.0**; HbA1c **≥48 mmol/mol (6.5%)**

Item	Value
DKA (ISPAD 2022)	Glucose **>11**, pH **<7.3** or HCO_3 **<18**, BOHB **≥3**
DKA severity	pH <7.3 mild; **<7.2** moderate; **<7.1** severe
Shock bolus	0.9% saline **20 mL/kg** (NICE 10 mL/kg)
Non-shock bolus	**10–20 mL/kg** over 20–60 min
Deficit	**5%** (pH ≥7.1); **7–10%** (pH <7.1); over **24–48 h**
Insulin infusion	**0.05–0.1 units/kg/h**, **1–2 h after fluids**, **no bolus**
Potassium	**40 mmol/L** (UCG 20–40) once K <5.5 and urine passed
Add dextrose	Glucose ≈**14–17 mmol/L** (UCG <14)
Glucose fall target	**2–5 mmol/L per hour**
Cerebral oedema	Mannitol **0.5–1 g/kg** or 3% saline **2.5–5 mL/kg** over 10–15 min
Corrected Na	Na + 0.3 × (glucose − 5.5)
Insulin TDD	Start ~**0.5** units/kg/day; prepubertal **0.7–1**; puberty **1–1.5**; premix **2/3 am, 1/3 pm**
HbA1c target	**<7% (53 mmol/mol)**
Levothyroxine (CH)	**10–15 µg/kg/day**, start within 2 weeks of birth
Short stature	Height Z **<−2**; MPH ± 6.5 cm
Puberty	Precocious: girls <8, boys <9; delayed: girls >13, boys >14
Adrenal crisis hydrocortisone	**50–100 mg/m²** IV (infant 25 mg; 1–5 y 50 mg; >5 y 100 mg)
CAH maintenance	Hydrocortisone **10–15 mg/m²/day** ÷ 3; fludrocortisone **0.05–0.2 mg/day**; NaCl **1–2 g/day** in infancy
Rickets	Vitamin D **2,000 IU/day** (<1 y) / **3,000–6,000 IU/day** (1–12 y) × ≥12 weeks + calcium **500 mg/day**
Hypocalcaemic seizure	10% calcium gluconate **0.5 mL/kg** (max 20 mL) IV over 5–10 min

Classic exam traps

- **"Give an insulin bolus of 0.1 units/kg to start DKA treatment."** Wrong: **no bolus**; infusion starts 1–2 h after fluids.
- **"Glucose is now 12 mmol/L, so stop the insulin."** Wrong: **add dextrose** and continue insulin until ketoacidosis resolves.
- **"Serum K is 5.0, so no potassium is needed."** Wrong: total-body K is low; add K once urine is passed.
- **"Sodium bicarbonate for pH 7.0."** Wrong: not routine; increases cerebral oedema risk.
- **"Drowsy, deep breathing, febrile child = cerebral malaria and pneumonia."** Check glucose and urine ketones: it may be **DKA**.
- **"Baby with ambiguous genitalia looks more like a boy, so register as male."** Wrong: **don't assign sex** until karyotype and assessment; check electrolytes.
- **"Short child: start with a growth hormone test."** Wrong: look for **malnutrition and chronic disease**, measure parents, check velocity, bone age, TSH, karyotype in girls.
- **"Rickets: vitamin D alone."** Often wrong in Africa: give **calcium** too.

Questions seniors ask

What is your fluid plan for this 20 kg child with moderate DKA (pH 7.15), not shocked? 0.9% saline 10 mL/kg (200 mL) over 1 hour; deficit 5% = 1,000 mL (minus the bolus per NICE) plus maintenance 1,500 mL/day over 48 h ≈ 80 mL/h; KCl 40 mmol/L once urine passed; insulin 1–2 units/h after 1–2 hours; dextrose at 14 mmol/L.

Why do we wait before starting insulin? Glucose falls with fluids alone; early insulin has been associated with cerebral oedema and causes sharp falls in potassium and osmolality.

The child's GCS has dropped from 15 to 12 at 6 hours and HR is 70 with BP 130/80. What now? Cerebral oedema: head up, reduce fluids by a third, mannitol 0.5–1 g/kg (or 3% saline 2.5–5 mL/kg) over 10–15 minutes, call ICU/anaesthesia; scan after treatment.

How would you diagnose congenital hypothyroidism without screening? Clinical suspicion (prolonged jaundice, hypotonia, constipation, large tongue, umbilical hernia, large posterior fontanelle) → TSH and FT4 → start levothyroxine immediately.

A 2-week-old boy with vomiting, Na 122, K 7.4 and glucose 2.1: diagnosis and immediate treatment? Salt-wasting adrenal crisis (CAH): 0.9% saline 20 mL/kg, 10% dextrose 2 mL/kg, IV hydrocortisone 25 mg, then 6-hourly; ECG and K monitoring.

How do you counsel a family about sick-day rules? Never stop insulin, check glucose and ketones often, give extra insulin for high ketones, keep drinking, and come to hospital early if vomiting, deep breathing or drowsy.

What causes rickets in a sunny country like Uganda? Low dietary calcium, infants kept indoors or fully covered, prolonged breastfeeding without complementary foods by vitamin D–deficient mothers, dark skin, malnutrition, chronic illness and anticonvulsants.

SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Diagnostic criteria for diabetes in a child?	Symptoms + **random glucose ≥11.1 mmol/L**; or **fasting ≥7.0 mmol/L**; or 2-h OGTT ≥11.1; or **HbA1c ≥48 mmol/mol (6.5%)**. Without symptoms, confirm on another day.
2	The 4 Ts of new type 1 diabetes?	**Toilet** (polyuria, nocturia, new bed-wetting), **Thirsty**, **Tired**, **Thinner** (weight loss).
3	ISPAD 2022 definition of DKA?	Glucose **>11 mmol/L** + venous **pH <7.3 or HCO_3 <18 mmol/L** + **ketonaemia (BOHB ≥3 mmol/L)** or moderate/large ketonuria.
4	DKA severity by pH and bicarbonate (ISPAD 2022)?	**Mild**: pH <7.3 or HCO_3 <18. **Moderate**: pH <7.2 or HCO_3 <10. **Severe**: pH <7.1 or HCO_3 <5 mmol/L. (NICE uses HCO_3 <15 for mild.)
5	Initial fluid in DKA?	**Shock**: 0.9% saline **20 mL/kg** rapidly (NICE: 10 mL/kg, discuss before more). **No shock**: **10–20 mL/kg** over 20–60 min (NICE 10 mL/kg over 60 min; UCG 10–20 mL/kg).
6	How do you estimate the DKA fluid deficit and over how long do you replace it?	Assume **5%** dehydration if pH ≥7.1 and **7–10%** if pH <7.1; give deficit **plus maintenance evenly over 24–48 h** (NICE: 48 h), subtracting a non-shock bolus (NICE).
7	Insulin dose and timing in paediatric DKA?	**Soluble insulin 0.05–0.1 units/kg/h IV**, started **1–2 hours after IV fluids begin**; **no bolus**. UCG 2023: 0.05 units/kg/h under 5 years, 0.1 over 5 years.
8	Potassium replacement in DKA?	**40 mmol/L** in IV fluids once K <5.5 and urine passed (start immediately if K <3.5). UCG: 20 mmol/L if K 3.5–5.5, 40 if <3.5, none if >5.5; 1 ampoule KCl 15% 10 mL = 20 mmol.
9	When do you add dextrose to DKA fluids?	When glucose falls to about **14–17 mmol/L** (UCG: <14 mmol/L) or falls faster than 5 mmol/L/h; **keep the insulin running**.
10	Signs of cerebral oedema in DKA?	Headache, recurrent vomiting, irritability/lethargy, **falling GCS**, **bradycardia with rising BP** (Cushing response), abnormal posturing, sluggish pupils, cranial nerve palsy, incontinence; usually 4–12 h into treatment.
11	Treatment of cerebral oedema in DKA?	Head up 30°, reduce fluids by a third, **mannitol 0.5–1 g/kg IV over 10–15 min** or **3% saline 2.5–5 mL/kg**; call ICU; scan only after treatment starts.
12	Why no bicarbonate in DKA?	No benefit for outcome, and it is associated with **cerebral oedema**, hypokalaemia and paradoxical CSF acidosis. Only for life-threatening hyperkalaemia or cardiovascular collapse.
13	Corrected sodium formula in DKA, and why track it?	**Corrected Na = Na + 0.3 × (glucose − 5.5)** mmol/L. A **falling** corrected Na during treatment is a warning sign of cerebral oedema.
14	Starting subcutaneous insulin after DKA (premixed regimen common in Uganda)?	Total daily dose about **0.5–1 unit/kg/day** (prepubertal 0.7–1; puberty 1–1.5; UCG: start 0.5 under 5 years), premixed 30/70 **2/3 before breakfast, 1/3 before supper**. Stop IV insulin 30–60 min after the first SC dose.
15	ISPAD 2022 glycaemic targets for children with T1DM?	**HbA1c <7% (53 mmol/mol)**; glucose mostly **4–10 mmol/L**; avoid hypoglycaemia.

#	Question	Answer
16	Sick-day rules for a child with T1DM?	**Never stop insulin**; check glucose 2–3 hourly and ketones; extra rapid-acting insulin (10–20% of TDD) if ketones high; keep drinking; come to hospital for vomiting, deep breathing, drowsiness or rising ketones.
17	Clinical features of congenital hypothyroidism?	Prolonged jaundice, poor feeding, sleepiness, constipation, hypotonia, **large tongue**, **umbilical hernia**, **large posterior fontanelle**, hoarse cry, dry skin, coarse facies, hypothermia. Often normal at birth.
18	Levothyroxine dose in congenital hypothyroidism?	**10–15 µg/kg/day** orally once daily (about 37.5–50 µg for a term baby), started **as soon as possible (ideally within 2 weeks of birth)**; crush and mix with breast milk; avoid giving with soy, iron or calcium.
19	Mid-parental height formula?	**(Father + mother) ÷ 2, +6.5 cm for boys, −6.5 cm for girls**; target range about ±8.5 cm.
20	Familial short stature vs constitutional delay?	**Familial**: short parents, normal velocity, **bone age = chronological age**, normal puberty timing. **Constitutional delay**: normal velocity, **delayed bone age**, late puberty, family history of late bloomers; reaches target height.
21	Why karyotype every short girl?	**Turner syndrome (45,X)** may present with short stature alone; diagnosis allows cardiac/renal screening, puberty induction and growth hormone (specialist).
22	Commonest cause of ambiguous genitalia in a 46,XX newborn?	**Congenital adrenal hyperplasia due to 21-hydroxylase deficiency** (90–95% of CAH): virilised external genitalia, normal uterus and ovaries.
23	Salt-wasting crisis in CAH: timing and biochemistry?	At **1–3 weeks** of age: vomiting, poor feeding, weight loss, dehydration, shock; **Na low, K high, glucose low**, metabolic acidosis. Boys may have normal genitalia (± pigmented scrotum).
24	Emergency hydrocortisone doses for adrenal crisis?	**50–100 mg/m² IV** bolus; practical: **infant 25 mg, 1–5 years 50 mg, >5 years 100 mg**; then 50–100 mg/m²/day divided 6-hourly; plus 0.9% saline 20 mL/kg and dextrose.
25	CAH maintenance treatment?	**Hydrocortisone 10–15 mg/m²/day** in 3 doses; **fludrocortisone 0.05–0.2 mg/day**; **NaCl 1–2 g/day** in infancy; **double/triple doses** when ill; IM hydrocortisone if vomiting.
26	Biochemistry of nutritional rickets?	**High ALP**, **low phosphate**, calcium low or normal, **high PTH**, **25-OH vitamin D <30 nmol/L** (deficiency). X-ray: cupping, splaying, fraying of metaphyses.
27	Treatment of nutritional rickets (Global Consensus 2016)?	Vitamin D **2,000 IU/day** (<1 year) or **3,000–6,000 IU/day** (1–12 years) for **≥12 weeks** (or stoss 50,000 / 150,000 IU), **plus calcium 500 mg/day**, then maintenance 400–600 IU/day. Recheck ALP and X-ray at 3 months.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 10-year-old girl (25 kg) presents to Mengo with vomiting, abdominal pain and deep sighing breathing. She was treated for malaria last week. GCS 14, HR 128, CRT 2 s, BP 104/64, chest clear, SpO_2 98%. Which bedside test is MOST likely to give the diagnosis?

- A Malaria rapid diagnostic test
- B Pulse oximetry
- C Capillary glucose and urine dipstick for ketones
- D Haemoglobin
- E Abdominal ultrasound

2. A 12-year-old boy (30 kg) has DKA: glucose 29 mmol/L, pH 7.15, HCO_3 8 mmol/L, ketones +++. He is not shocked. He has received 10 mL/kg 0.9% saline over 1 hour. What is the MOST appropriate insulin plan?

- A Soluble insulin 3 units IV bolus followed by 3 units/h
- B Premixed insulin 30/70 15 units SC now
- C Wait until the glucose is below 14 mmol/L before starting insulin
- D Soluble insulin 0.5 units/kg/h IV
- E Soluble insulin infusion 0.05–0.1 units/kg/h (1.5–3 units/h) IV, no bolus

3. A 6-year-old (20 kg) with DKA has been on insulin 0.1 units/kg/h for 4 hours. Glucose has fallen from 30 to 13 mmol/L; pH 7.18, ketones still +++. What is the BEST next step?

- A Add 5% dextrose to the IV fluids and continue the insulin infusion
- B Stop the insulin infusion until glucose rises
- C Reduce insulin to 0.01 units/kg/h
- D Give 10% dextrose 5 mL/kg as a bolus
- E Switch to SC premixed insulin

4. Six hours into DKA treatment, a 4-year-old (15 kg) becomes irritable, complains of headache and vomits. GCS falls from 15 to 11. HR 72 (was 140), BP 128/84. Glucose 12 mmol/L. What should you do FIRST? Advanced

- A Arrange an urgent CT scan before any treatment
- B Give mannitol 0.5–1 g/kg IV over 10–15 minutes, raise the head, reduce fluids by a third and call ICU
- C Give a 20 mL/kg bolus of 0.9% saline
- D Give 10% dextrose 5 mL/kg
- E Increase the insulin infusion to 0.2 units/kg/h

5. A 9-year-old with DKA has initial K 5.0 mmol/L and is passing urine. Which potassium plan is MOST appropriate?

- A No potassium because the level is normal
- B Give potassium only if the ECG shows U waves
- C Oral potassium once she is eating
- D Add potassium chloride 40 mmol/L to the IV fluids and monitor K every 2–4 hours
- E Give KCl 1 mmol/kg as an IV push

6. A 4-month-old girl is brought for poor weight gain. She was jaundiced until 6 weeks, sleeps a lot, feeds slowly and passes stool every 4–5 days. She has a large tongue, an umbilical hernia, dry skin and a posterior fontanelle of 1.5 cm. Which is the MOST appropriate next step?

A Reassure; these are normal variants
B Start oral iron for anaemia
C Refer for umbilical hernia repair
D Check TSH and free T4 and start levothyroxine 10–15 µg/kg/day if confirmed, without delay
E Give lactulose for constipation and review in 3 months

7. A 12-day-old boy (birth weight 3.3 kg, now 2.9 kg) has vomited for 3 days. He is lethargic, CRT 4 s, scrotum looks pigmented. Na 121 mmol/L, K 7.3 mmol/L, glucose 2.0 mmol/L. Which is the MOST important specific treatment along with saline and dextrose? Advanced

A Pyloromyotomy
B IV hydrocortisone 25 mg, then 6-hourly
C Furosemide for hyperkalaemia
D IV ceftriaxone alone
E Oral rehydration solution

8. At Mengo, a term baby is born with clitoromegaly (phallus 1.8 cm), fused labioscrotal folds, a single perineal opening and no palpable gonads. The father asks whether it is a boy or a girl. What is the BEST response and plan?

A Tell him it is a boy because the phallus is large
B Tell him it is a girl and discharge tomorrow
C Advise circumcision before discharge
D Wait until puberty to decide
E Explain that the baby's genitals have not finished developing typically, that tests will guide the decision, delay registration of sex, check glucose and electrolytes daily, and arrange ultrasound, karyotype and 17-OHP

9. An 11-year-old boy is 129 cm (HAZ −2.2). Father 165 cm, mother 154 cm. His health records show steady growth along the same line at 5 cm/year. Bone age is 9 years; his father started puberty at 16. Examination is normal. What is the MOST likely diagnosis?

A Constitutional delay of growth and puberty
B Growth hormone deficiency
C Hypothyroidism
D Turner syndrome
E Achondroplasia

10. A 3-year-old boy (11 kg) has bowed legs, wide wrists, a rachitic rosary and delayed walking. He was breastfed to 2 years with little milk, eggs or fish in his diet. ALP 980 IU/L, phosphate low, calcium 2.1 mmol/L. Which treatment is BEST?

A Calcium alone for 2 weeks
B Vitamin D 400 IU/day indefinitely
C Vitamin D 3,000–6,000 IU/day for at least 12 weeks plus calcium 500 mg/day, and dietary advice
D Refer for osteotomy now
E Phosphate supplements and calcitriol for X-linked hypophosphataemic rickets

11. A 14-year-old with known T1DM on premixed insulin presents with 2 days of vomiting and diarrhoea. His mother stopped his insulin because he was not eating. Glucose 22 mmol/L, urine ketones +++, pH 7.26. Which advice would have MOST likely prevented this admission?

A Stop insulin whenever the child cannot eat

B Halve all doses during any illness

C Give oral metformin instead of insulin during illness

D Rely on urine glucose testing only

E Never stop insulin during illness; check glucose and ketones often and give extra rapid-acting insulin if ketones are high, with fluids

12. A 25 kg child with severe DKA (pH 7.02), not shocked, received 0.9% saline 250 mL over 1 hour. Using NICE (48-hour replacement, 10% deficit, subtracting the non-shock bolus), approximately what hourly rate should the fluid run at? (Maintenance for 25 kg = 1,600 mL/day.) Advanced

A About 50 mL/h

B About 80 mL/h

C About 115 mL/h

D About 200 mL/h

E About 300 mL/h

13. A 13-year-old girl is 136 cm (HAZ −2.6) with no breast development. She has a low hairline, wide carrying angle and widely spaced nipples. Her parents are of average height. Which investigation is MOST important?

A Karyotype

B Growth hormone stimulation test

C Serum vitamin D

D Bone age only, then reassure

E Coeliac serology

14. A Ugandan HC IV has no infusion pump. A 7-year-old (20 kg) with moderate DKA has been rehydrated for 1 hour with 0.9% saline. Which insulin approach is MOST appropriate, consistent with UCG 2023 and ISPAD? Advanced

A Premixed 30/70 insulin 20 units SC once

B Insulin 20 units IV push once

C No insulin until transfer, even if it takes 24 hours

D Soluble insulin 0.1 units/kg (2 units) IM or SC every hour, with hourly glucose and dextrose added when glucose falls to about 14 mmol/L

E Oral glibenclamide

15. A 7-year-old with CAH on hydrocortisone 5 mg three times daily develops fever 39 °C and cough but is eating and not vomiting. What is the MOST appropriate advice?

A Continue the usual dose; fever does not matter

B Double or triple the hydrocortisone dose until well, and give IM hydrocortisone and come to hospital if vomiting or drowsy

C Stop hydrocortisone while febrile

D Switch to fludrocortisone alone

E Give dexamethasone 0.6 mg/kg once

Answers & explanations

1. C · Capillary glucose and urine dipstick for ketones

C is correct: Kussmaul breathing with a clear chest, vomiting and abdominal pain suggest **DKA**; a glucometer reading (often 'HI') and **urine ketones** confirm it in minutes. A is wrong: she was recently treated and malaria would not explain acidotic breathing with a clear chest as well. B is wrong: SpO_2 is normal; the problem is metabolic, not pulmonary. D is wrong: Hb will not distinguish the cause. E is wrong: DKA abdominal pain is common and ultrasound would delay diagnosis.

2. E · Soluble insulin infusion 0.05–0.1 units/kg/h (1.5–3 units/h) IV, no bolus

E is correct: ISPAD 2022/NICE give **0.05–0.1 units/kg/h** soluble insulin by infusion, started 1–2 h after fluids, **without a bolus**. A is wrong: a bolus increases the risk of hypokalaemia and cerebral oedema without speeding recovery. B is wrong: SC premixed insulin is for after DKA resolves. C is wrong: insulin is needed to stop ketone production regardless of glucose; dextrose is added as glucose falls. D is wrong: 0.5 units/kg/h is five to ten times the correct dose.

3. A · Add 5% dextrose to the IV fluids and continue the insulin infusion

A is correct: glucose has reached about 14 mmol/L but **ketoacidosis persists**; add **5% dextrose** and keep insulin running so ketones are cleared. B is wrong: stopping insulin restarts ketogenesis. C is wrong: such a low dose will not suppress ketones; ISPAD keeps at least 0.05 units/kg/h. D is wrong: she is not hypoglycaemic. E is wrong: SC insulin starts only when acidosis has resolved and she can eat.

4. B · Give mannitol 0.5–1 g/kg IV over 10–15 minutes, raise the head, reduce fluids by a third and call ICU

B is correct: headache, falling GCS and **bradycardia with hypertension** during DKA treatment are **cerebral oedema**; treat immediately with mannitol (or 3% saline 2.5–5 mL/kg), head elevation and fluid reduction. A is wrong: imaging must not delay treatment. C is wrong: more fluid may worsen oedema. D is wrong: glucose is 12 mmol/L; this is not hypoglycaemia. E is wrong: more insulin speeds osmolar shifts and does not treat oedema.

5. D · Add potassium chloride 40 mmol/L to the IV fluids and monitor K every 2–4 hours

D is correct: total-body potassium is depleted in DKA, and insulin will drive K into cells; ISPAD adds **40 mmol/L** once K <5.5 and urine is passed (UCG 20 mmol/L for K 3.5–5.5), with 2–4 hourly checks. A is wrong: a 'normal' serum K hides a deficit and will fall. B is wrong: waiting for ECG changes is dangerous. C is wrong: replacement is needed during IV treatment, not after. E is wrong: IV push potassium can cause fatal arrhythmia.

6. D · Check TSH and free T4 and start levothyroxine 10–15 µg/kg/day if confirmed, without delay

D is correct: prolonged jaundice, lethargy, constipation, macroglossia, umbilical hernia and a large posterior fontanelle are classic **congenital hypothyroidism**; urgent TSH/FT4 and **levothyroxine 10–15 µg/kg/day** protect development. A is wrong: the combination is not normal. B is wrong: nothing suggests iron deficiency as the main problem. C is wrong: the hernia is a sign, and most close spontaneously. E is wrong: treating the constipation alone delays diagnosis while brain damage accumulates.

7. B · IV hydrocortisone 25 mg, then 6-hourly

B is correct: hyponatraemia, hyperkalaemia, hypoglycaemia and shock at 1–3 weeks (with pigmentation) = **salt-wasting CAH adrenal crisis**; give **IV hydrocortisone** (infant 25 mg) with 0.9% saline 20 mL/kg and dextrose. A is wrong: pyloric stenosis causes hypochloraemic hypokalaemic alkalosis, not hyperkalaemia. C is wrong: furosemide worsens volume depletion. D is wrong: sepsis cover is reasonable but will not correct cortisol deficiency. E is wrong: a shocked infant needs IV treatment.

8. E · Explain that the baby's genitals have not finished developing typically, that tests will guide the decision, delay registration of sex, check glucose and electrolytes daily, and arrange ultrasound, karyotype and 17-OHP

E is correct: ambiguous genitalia with **impalpable gonads** is most likely **46,XX CAH**; sex should not be assigned until assessment, and the baby must be watched for **salt-wasting** with daily electrolytes and glucose while ultrasound, karyotype and 17-OHP are done. A and B are wrong: guessing the sex can lead to wrong assignment and a missed adrenal crisis. C is wrong: circumcision must be avoided (tissue may be needed for reconstruction). D is wrong: sex of rearing is decided in infancy by a multidisciplinary team with the family.

9. A · Constitutional delay of growth and puberty

A is correct: **normal velocity**, **delayed bone age**, normal examination and a father who was a late developer = **constitutional delay**; he will catch up later. B is wrong: GH deficiency gives a falling velocity. C is wrong: hypothyroidism also causes falling velocity and other signs. D is wrong: Turner syndrome affects girls. E is wrong: achondroplasia causes disproportionate short stature evident from infancy.

10. C · Vitamin D 3,000–6,000 IU/day for at least 12 weeks plus calcium 500 mg/day, and dietary advice

C is correct: this is **nutritional rickets** (vitamin D and/or calcium deficiency); the Global Consensus gives **3,000–6,000 IU/day** (1–12 years) for ≥12 weeks **with calcium 500 mg/day**, then maintenance. A is wrong: calcium alone for 2 weeks is too little and too short. B is wrong: 400 IU/day is the prevention/maintenance dose. D is wrong: deformities often improve with treatment; surgery is considered only after healing. E is wrong: XLH has very low phosphate with normal calcium and a family history; diet history here points to nutritional rickets.

11. E · Never stop insulin during illness; check glucose and ketones often and give extra rapid-acting insulin if ketones are high, with fluids

E is correct: **sick-day rules**: insulin is needed even when not eating because illness raises counter-regulatory hormones; check glucose and ketones and give **extra insulin** for ketosis. A is wrong: stopping insulin is the classic cause of DKA in known diabetics. B is wrong: needs usually rise in illness. C is wrong: metformin cannot replace insulin in T1DM. D is wrong: urine glucose does not detect ketosis or hypoglycaemia.

12. C · About 115 mL/h

C is correct: deficit 10% × 25 kg = 2,500 mL − 250 mL bolus = 2,250 mL; maintenance 1,600 × 2 = 3,200 mL; total 5,450 mL ÷ 48 h ≈ **114 mL/h**. A is wrong: that would barely cover maintenance. B is wrong: this ignores most of the deficit. D and E are wrong: rates this high risk fluid overload and correct the deficit far faster than recommended.

13. A · Karyotype

A is correct: short stature with **absent puberty** and **dysmorphic features** in a girl strongly suggests **Turner syndrome (45,X)**; a karyotype confirms it and triggers cardiac (coarctation, bicuspid valve) and renal screening and puberty induction. B is wrong: GH testing is not the first step and Turner girls are not GH deficient. C is wrong: nothing suggests rickets. D is wrong: reassurance would miss the diagnosis. E is wrong: coeliac disease is a cause of short stature but not of these dysmorphic features.

14. D · Soluble insulin 0.1 units/kg (2 units) IM or SC every hour, with hourly glucose and dextrose added when glucose falls to about 14 mmol/L

D is correct: where an IV infusion cannot be run safely, **hourly IM/SC soluble (or rapid-acting) insulin 0.1 units/kg** is an accepted alternative (UCG lists hourly IM insulin; ISPAD accepts SC regimens) once perfusion is restored, with hourly glucose and dextrose. A is wrong: a single large premixed dose gives unpredictable absorption and hypoglycaemia. B is wrong: a 1 unit/kg IV push is a dangerous overdose. C is wrong: withholding insulin allows ketoacidosis to progress. E is wrong: sulphonylureas do not work in T1DM.

15. B · Double or triple the hydrocortisone dose until well, and give IM hydrocortisone and come to hospital if vomiting or drowsy

B is correct: **stress dosing** (double or triple oral hydrocortisone) during febrile illness prevents an adrenal crisis; if vomiting or unwell, IM hydrocortisone and hospital. A is wrong: illness increases cortisol needs. C is wrong: stopping steroids risks an adrenal crisis. D is wrong: fludrocortisone replaces aldosterone, not cortisol. E is wrong: a single dexamethasone dose is not the standard stress-dose plan and does not replace mineralocorticoid needs.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Severe DKA misdiagnosed as malaria

A 9-year-old girl is brought to Mengo emergency at 20:00. She was treated for malaria at a clinic 3 days ago but has kept vomiting, has abdominal pain and is now drowsy with fast deep breathing. Her mother says she has been drinking a lot and wetting the bed for the past month. Weight 24 kg (she was 27 kg at school a month ago).

1. What is your working diagnosis and your first bedside tests?

> **DKA** (new type 1 diabetes): polyuria, polydipsia, weight loss, vomiting, Kussmaul breathing, drowsiness. Bedside: **glucometer** ('HI' → lab 31 mmol/L), **urine ketones** (+++), mRDT (negative), Hb, SpO_2 (98%), GCS (13).

2. Assess her severity and haemodynamic state. Which bloods do you send?

> HR 132, CRT 3 s, warm peripheries, BP 98/60: **dehydrated but not shocked**. Venous gas **pH 7.06, HCO_3 5 → severe DKA**. Electrolytes: Na 131, K 5.2, urea 9; FBC; blood culture if febrile; HbA1c. Corrected Na = 131 + 0.3 × (31 − 5.5) ≈ **139**.

3. Prescribe her fluids for the first 48 hours.

> Bolus **0.9% saline 10 mL/kg = 240 mL over 1 h** (UCG 10–20 mL/kg). Deficit **10%** = 2,400 mL − 240 = 2,160 mL; maintenance 1,580 mL/day × 2 = 3,160 mL; total **5,320 mL over 48 h ≈ 110 mL/h** of 0.9% saline with **KCl 40 mmol/L** once urine passed (K 5.2 <5.5).

4. When and how do you start insulin?

> **After 1–2 h of fluids**: soluble insulin **0.1 units/kg/h = 2.4 units/h** (50 units in 50 mL 0.9% saline → 2.4 mL/h) by syringe pump; **no bolus**. If no pump: 0.1 units/kg (2.4 units) IM/SC hourly. Add **5% dextrose** at glucose ≈14 mmol/L; aim for a fall of 2–5 mmol/L/h.

5. At 7 hours she complains of headache and her GCS falls to 11, HR 68, BP 130/85. What do you do?

> **Cerebral oedema**: head up 30°, **mannitol 0.5–1 g/kg = 12–24 g (60–120 mL of 20%) IV over 10–15 min** (or 3% saline 60–120 mL), reduce fluids by a third, call senior/ICU, prepare for airway support; CT after treatment if available.

6. She recovers. What must happen before discharge?

> Resolution (pH >7.3, HCO_3 >18, ketones clearing, eating) → **SC insulin** (e.g. premixed 30/70 about 0.7–0.8 units/kg/day = ~18 units/day: 12 units am, 6 units pm, or basal-bolus) with the first SC dose 30–60 min before stopping the infusion; **education** for mother and child (injections, storage, glucose testing, hypoglycaemia, sick-day rules, diet, school letter); glucometer and insulin supply via the diabetes clinic; follow-up in 1–2 weeks.

Take-away: Polyuria + vomiting + deep breathing = check glucose and ketones; in DKA give fluids first, insulin after 1–2 hours without a bolus, add potassium and then dextrose, and watch the brain.

Case 2: The floppy, jaundiced baby

A 3-month-old boy is brought to the young child clinic because 'he is too quiet'. He was born at term, 3.6 kg, at a HC III. He was jaundiced until 7 weeks, feeds slowly, sleeps most of the day and passes hard stools twice a week. Weight now 4.6 kg.

1. What features do you look for on examination?

> Coarse facies, **large tongue**, hoarse cry, **umbilical hernia**, **large posterior fontanelle**, hypotonia, dry mottled skin, cool peripheries, slow pulse, goitre, abdominal distension, poor weight gain (WAZ falling). He has macroglossia, umbilical hernia, posterior fontanelle 1.5 cm, hypotonia, HR 96.

2. What is the likely diagnosis and how would you confirm it?

> **Congenital hypothyroidism** (no newborn screening in Uganda). Send **TSH and free T4** (TSH 160 mU/L, FT4 very low). Thyroid ultrasound later (absent/ectopic gland vs goitre). Exclude other causes of prolonged jaundice (conjugated bilirubin, stool colour).

3. How will you treat him?

> **Levothyroxine 10–15 µg/kg/day**: 4.6 kg × 10–15 = 46–69 µg → **50 µg once daily**, crushed in a little breast milk by spoon, at the same time daily, not with soy formula, iron or calcium. Start the same day; do not wait for scans.

4. How will you monitor him?

> TSH and FT4 at **2 and 4 weeks**, then every 1–2 months in the first 6 months and every 3–4 months to age 3; adjust dose to keep FT4 in the upper half of normal and TSH normal; plot weight, length and head circumference; developmental assessment at each visit.

5. What do you tell the parents about the outlook?

> Treatment is lifelong (reassess at 3 years if transient is suspected) and must never be missed. Because diagnosis was at 3 months, there may be some developmental delay; early stimulation, follow-up and therapy help. Future babies should be checked early (dyshormonogenesis is recessive).

Take-away: Without newborn screening, clinical suspicion (prolonged jaundice, constipation, hypotonia, large tongue, umbilical hernia) is the screen; start levothyroxine 10–15 µg/kg/day the same day.

Case 3: Ambiguous genitalia and a crisis at 2 weeks

A baby is born at Mengo at term, 3.2 kg, with a 2 cm phallus, fused rugose labioscrotal folds, a single perineal opening and no palpable gonads. The parents ask the midwife whether the baby is a boy or a girl. The baby is discharged on day 2 'as a boy' by the family and returns on day 14 vomiting and floppy.

1. What should have happened at birth?

> **Do not assign sex**; explain sensitively that tests are needed; avoid circumcision; examine for gonads and pigmentation; **pelvic ultrasound** (uterus present?), **karyotype**, **17-OHP** after 24–48 h; **daily weight, glucose and electrolytes** for at least 2 weeks; refer to paediatric endocrinology/surgery; counselling.

2. On day 14: weight 2.8 kg, CRT 4 s, lethargic. Na 119, K 7.8, glucose 1.9 mmol/L. What is happening?

> **Salt-wasting adrenal crisis** from **21-hydroxylase CAH** (almost certainly a **46,XX virilised girl**): cortisol and aldosterone deficiency → hyponatraemia, hyperkalaemia, hypoglycaemia, hypovolaemic shock.

3. Give the emergency treatment with doses.

> **0.9% saline 20 mL/kg = 56 mL** rapidly (repeat if shocked); **10% dextrose 2 mL/kg = 5.6 mL** then dextrose-saline; **hydrocortisone 25 mg IV** then 50–100 mg/m²/day IV (BSA ≈0.2 m² → 10–20 mg/day, about 2.5–5 mg 6-hourly); **ECG**: if K causes arrhythmia, **10% calcium gluconate 0.5 mL/kg IV** over 5–10 min; antibiotics if sepsis possible; take cortisol/17-OHP before hydrocortisone only if it does not delay it.

4. Ultrasound shows a uterus; karyotype 46,XX; 17-OHP very high. What is the long-term treatment?

> **Hydrocortisone 10–15 mg/m²/day** in 3 doses, **fludrocortisone 0.05–0.2 mg/day**, **salt 1–2 g/day** in infancy; **stress dosing** (double/triple) when ill and IM hydrocortisone for vomiting; emergency card; growth and BP monitoring; genital surgery discussion with the specialist team.

5. How do you approach the family about sex of rearing?

> Honest, unhurried, multidisciplinary discussion (endocrinologist, surgeon, psychologist, family): **46,XX CAH girls have a uterus and ovaries and are fertile**, so most are raised as girls; acknowledge cultural and community pressures; support the parents to correct the registration; offer genetic counselling (25% recurrence risk).

Take-away: Never assign sex to a baby with ambiguous genitalia and impalpable gonads until assessed; watch electrolytes for 2–3 weeks, and treat a salt-wasting crisis with saline, dextrose and hydrocortisone.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.